



BAZAARO HYPERLOCAL MARKETPLACE DEPLOYMENT REPORT

SE3040 – Application Frameworks

1. Project Overview

Bazaaro is a full-stack hyperlocal marketplace platform enabling users to browse and purchase products and book local services. The system is built with a React frontend, an Express.js REST API backend, and MongoDB for persistent data storage. This deployment report covers the production release of version v1.0.0-beta submitted for Evaluation 02.

Module	SE3040 – Application Frameworks
Project Name	Bazaaro – Hyperlocal Marketplace for Products & Services
Version	v1.0.0-beta
Frontend URL	https://www.bazaro.online
Backend URL	https://api.bazaro.online
API Docs	https://api.bazaro.online/api-docs
Repository	https://github.com/kiruluchamika/Hyperlocal-Marketplace-for-Products-Services-Project

2. Deployment Architecture

The deployment uses separate cloud platforms for frontend and backend to allow independent scaling and operational control. MongoDB Atlas serves as the managed cloud database.

Layer	Platform	Domain / URL	Purpose
Frontend	Vercel	https://www.bazaro.online	React UI, routing, user/admin interaction
Backend	Render (Web Service)	https://api.bazaro.online	REST API, auth, business logic
Database	MongoDB Atlas	Cloud-managed cluster	Persistent data store
DNS	Custom Domain	bazaro.online	Root & www → Vercel, api → Render

High-Level Request Flow

- Browser requests the React frontend from Vercel (www.bazaro.online)
- React app sends API requests to api.bazaro.online (Render backend)
- Backend performs business logic and reads/writes data in MongoDB Atlas
- Stripe and other third-party services are called by the backend as needed
- Responses are returned through the same chain back to the browser

3. Backend Deployment Details

Platform & Service Configuration

Field	Value
Platform	Render
Service Type	Web Service
Runtime	Node.js
Repository	Hyperlocal-Marketplace-for-Products-Services-Project
Branch	main
Root Directory	backend
Build Command	npm install && npm run build
Start Command	npm start
Live URL	https://api.bazaro.online
API Docs URL	https://api.bazaro.online/api-docs
Health Endpoint	https://api.bazaro.online/api/settings

Setup Notes

- Do not hardcode the PORT variable — Render injects PORT at runtime
- All required backend environment variables must be set in Render's environment dashboard
- MongoDB Atlas network access must be configured to allow connections from Render's IP range
- Cold-start delays are expected on the free tier after periods of inactivity (Render spins down idle services)

Backend Environment Variables

Variable Name	Purpose
PORT	Injected by Render at runtime
MONGODB_URI	Primary MongoDB Atlas connection string
MONGODB_FALLBACK_URI	Optional fallback connection string
JWT_SECRET	Secret key for signing JSON Web Tokens
JWT_EXPIRES_IN	JWT expiry duration (e.g., 7d)
GOOGLE_CLIENT_ID	OAuth 2.0 client ID for Google login (optional)
TWILIO_ACCOUNT_SID	Twilio account identifier for OTP (optional)
TWILIO_AUTH_TOKEN	Twilio authentication token (optional)
TWILIO_VERIFY_SERVICE_SID	Twilio Verify service SID (optional)
STRIPE_SECRET_KEY	Stripe secret key for payment processing

STRIPE_PUBLISHABLE_KEY	Stripe publishable key
STRIPE_WEBHOOK_SECRET	Stripe webhook signing secret
STRIPE_CONNECT_ENABLED	Toggle for Stripe Connect feature
STRIPE_CONNECT_RETURN_URL	Return URL after Stripe Connect onboarding
STRIPE_CONNECT_REFRESH_URL	Refresh URL for Stripe Connect
STRIPE_PAYMENT_CURRENCY	Currency code for payments (e.g., LKR)
STRIPE_TRANSFER_FEE_PERCENT	Platform fee percentage for transfers
STRIPE_BALANCE_TO_LKR_RATE	Exchange rate used for LKR conversion
ENABLE_OTP_DELIVERY	Enable/disable OTP delivery
OTP_EXPIRY_MINUTES	OTP validity period in minutes
CURRENCY	Default app currency code
SMTP_HOST	SMTP server hostname (optional)
SMTP_PORT	SMTP server port
SMTP_SECURE	Use TLS for SMTP (true/false)
SMTP_USER	SMTP authentication username (optional)
SMTP_PASS	SMTP authentication password (optional)
SMTP_FROM_EMAIL	Sender email address
SMTP_FROM_NAME	Sender display name

4. Frontend Deployment Details

Platform & Build Configuration

Field	Value
Platform	Vercel
Framework Preset	Vite + React
Root Directory	frontend
Build Command	npm run build
Output Directory	dist
Routing	SPA fallback rewrite configured for direct route access
Live URL	https://www.bazaro.online
API Base URL in Prod	https://api.bazaro.online

Frontend Environment Variables

Variable Name	Purpose
VITE_GOOGLE_CLIENT_ID	Google OAuth Client ID for browser-side login
VITE_SOCKET_URL	WebSocket/Socket.IO server URL
VITE_STRIPE_PUBLISHABLE_KEY	Stripe publishable key for frontend payment UI

5. Environment Variables Summary

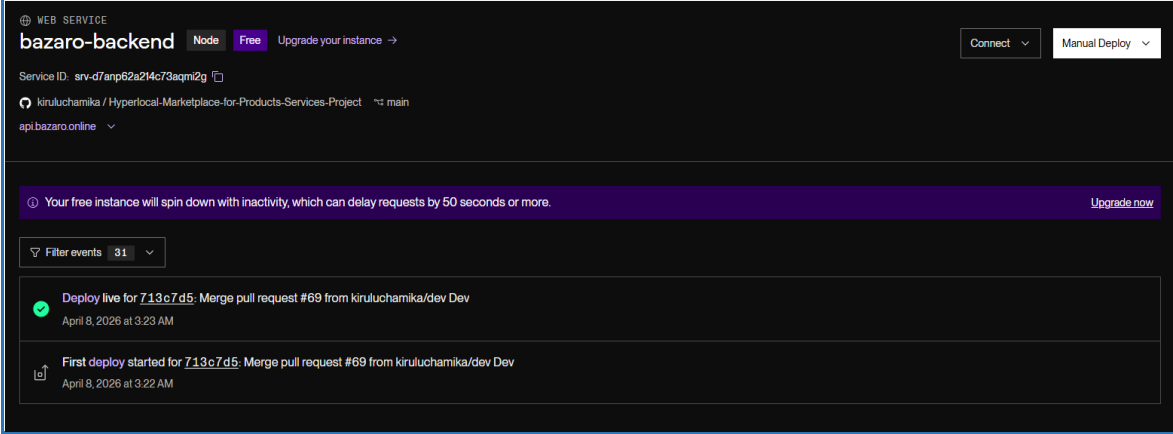
Environment Variable	Layer	Purpose	Example Format
MONGODB_URI	Backend	Database connection	mongodb+srv://user:****@cluster/db
JWT_SECRET	Backend	Token signing key	random-256bit-secret
STRIPE_SECRET_KEY	Backend	Payment processing	sk_live_****
STRIPE_WEBHOOK_SECRET	Backend	Webhook validation	whsec_****

6. Deployment Evidence

The following screenshots must be captured and included in this section to satisfy the deployment documentation requirement. Placeholders indicate where each screenshot should be inserted.

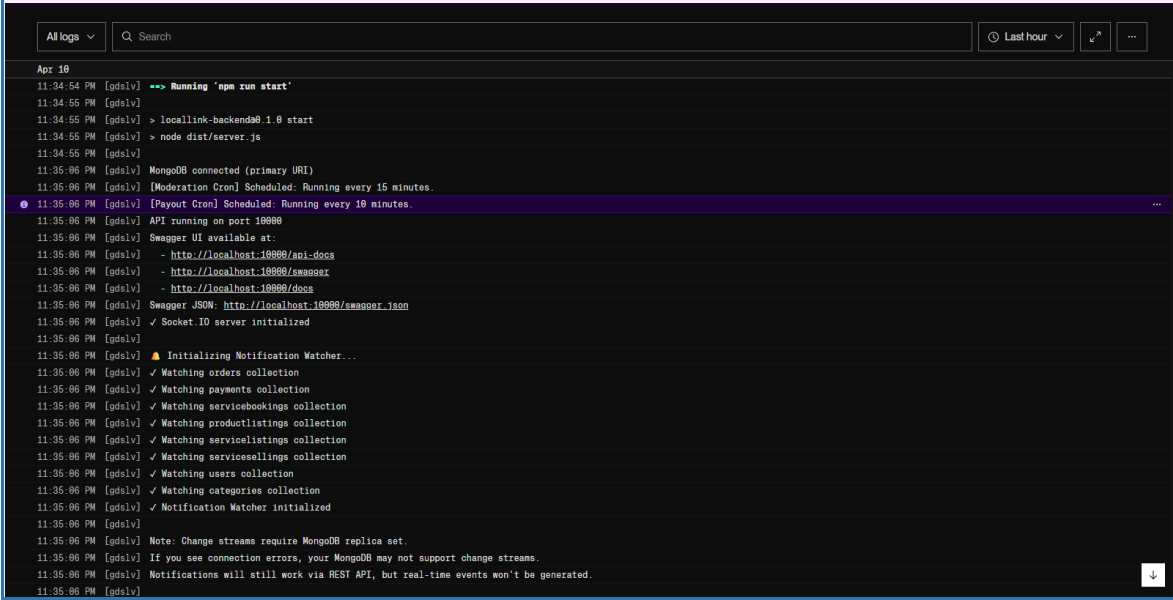
1. Render Service Overview

Screenshot of the Render dashboard showing the web service with a successful deploy status badge.



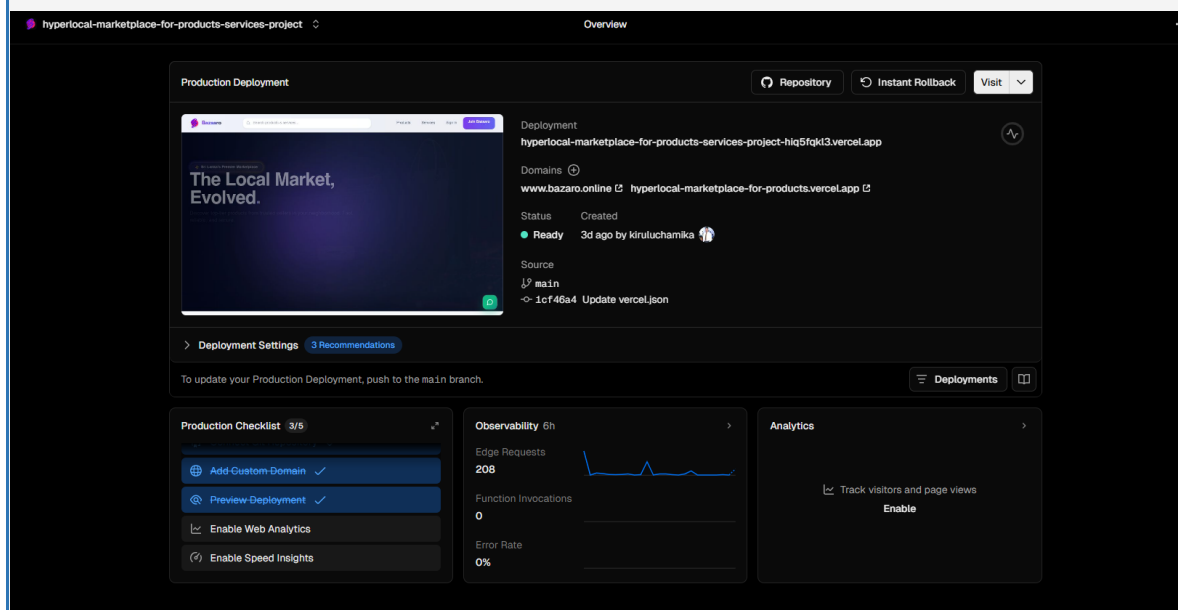
2. Render Deploy Logs

Screenshot of Render logs confirming 'Server started successfully' message at runtime.



3. Vercel Deployment Success

Screenshot of the Vercel project dashboard showing the latest production deployment as 'Ready'.



4. GitHub Release Page

Screenshot of the GitHub releases page showing the v1.0.0-beta tag and release notes.

The screenshot shows the GitHub release page for 'Bazaar Marketplace v1.0.0 Beta'. The page is dark-themed. At the top, the release title 'Bazaar Marketplace v1.0.0 Beta' is displayed with a 'Pre-release' badge. Below the title, it says 'kiruluchamika released this 8 hours ago' followed by the tag 'v1.0.0-beta' and the commit hash '1cf46a4'. The release description states: 'Beta release of Bazaar, a full-stack hyperlocal marketplace for products and services. This release is intended for testing and validation before the stable production release.' The 'Features' section lists: Product listings and service bookings, User and admin dashboards, Stripe payment integration, Real-time notifications, Maintenance mode and runtime payment controls, Admin settings panel for managing site behavior, and Vercel frontend deployment with Render backend support. The 'Notes' section lists: This is a beta release and may contain bugs or unfinished behavior, Some features or UI flows may still change before the stable release, and Feedback from testing is welcome. The 'Tech Stack' section lists: Frontend: React, Vite, TypeScript, Tailwind CSS; Backend: Node.js, Express, TypeScript, MongoDB; and Integrations: Stripe, Socket.IO, JWT. The 'Deployment' section lists: Frontend hosted on Vercel, Backend hosted on Render, and Custom domain configured with bazaro.online. At the bottom, the 'Assets' section shows two files: 'Source code (zip)' and 'Source code (tar.gz)', both uploaded 3 days ago.

Releases / v1.0.0-beta

Bazaar Marketplace v1.0.0 Beta Pre-release

kiruluchamika released this 8 hours ago v1.0.0-beta 1cf46a4

Bazaar Marketplace v1.0.0-beta

Beta release of Bazaar, a full-stack hyperlocal marketplace for products and services.

This release is intended for testing and validation before the stable production release.

Features

- Product listings and service bookings
- User and admin dashboards
- Stripe payment integration
- Real-time notifications
- Maintenance mode and runtime payment controls
- Admin settings panel for managing site behavior
- Vercel frontend deployment with Render backend support

Notes

- This is a beta release and may contain bugs or unfinished behavior
- Some features or UI flows may still change before the stable release
- Feedback from testing is welcome

Tech Stack

- Frontend: React, Vite, TypeScript, Tailwind CSS
- Backend: Node.js, Express, TypeScript, MongoDB
- Integrations: Stripe, Socket.IO, JWT

Deployment

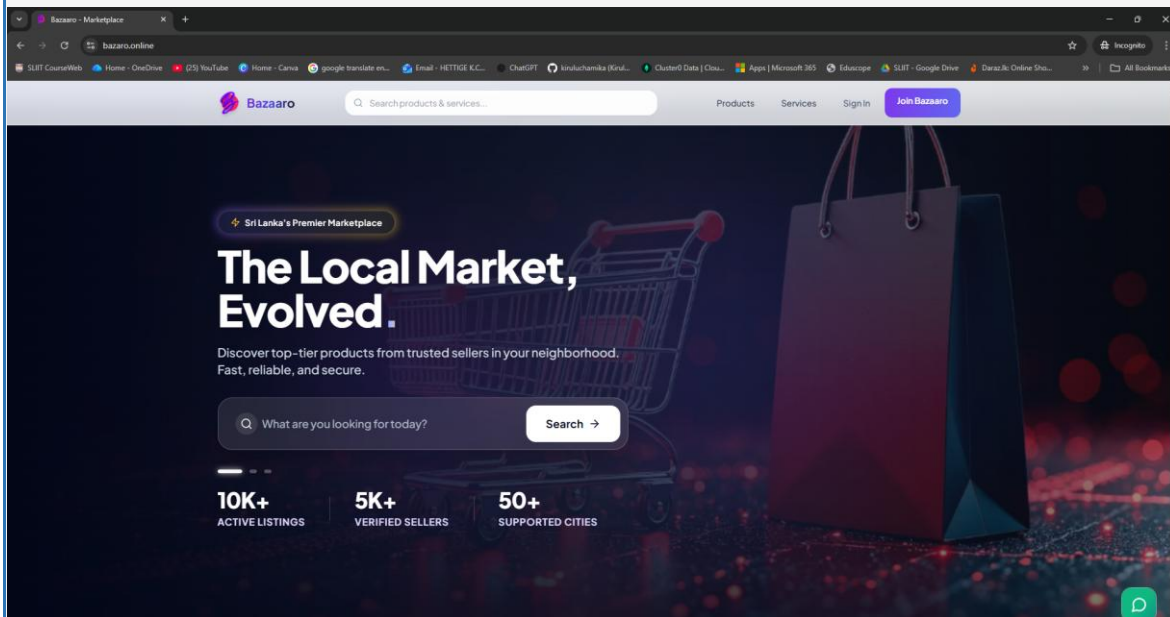
- Frontend hosted on Vercel
- Backend hosted on Render
- Custom domain configured with bazaro.online

Assets 2

- Source code (zip) 3 days ago
- Source code (tar.gz) 3 days ago

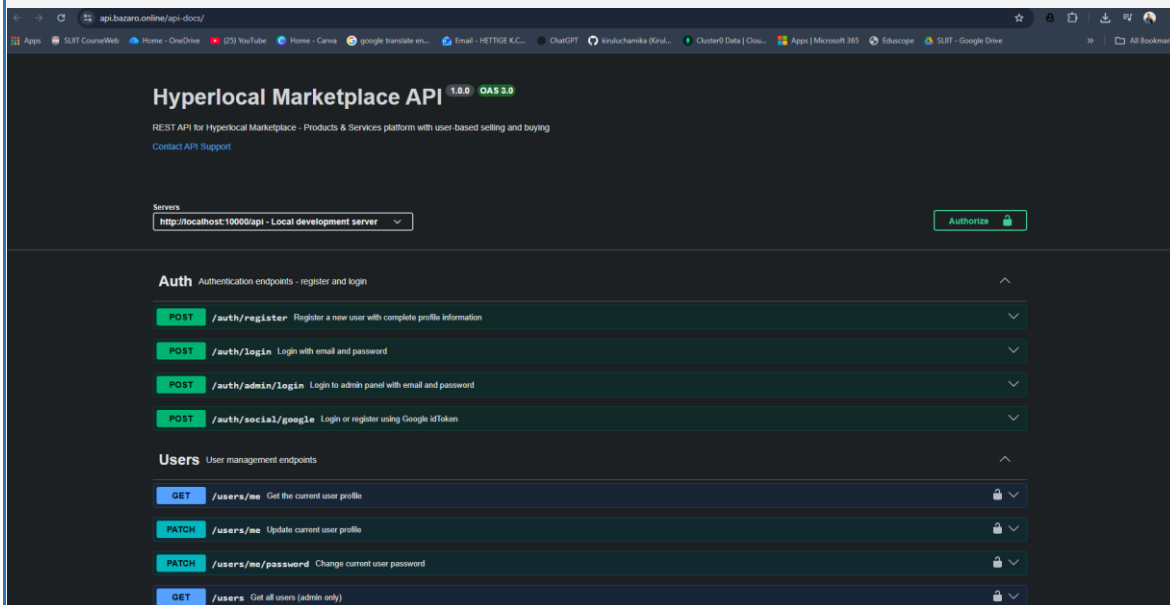
5. Live Frontend – Home Page

Screenshot of <https://www.bazaro.online> loaded in a browser, showing the home page with product/service listings.



6. Live Backend – API Docs

Screenshot of <https://api.bazaro.online/api-docs> showing the Swagger UI with all documented endpoints.



7. DNS Records

Screenshot of DNS provider showing A/CNAME records mapping `bazaro.online` → Vercel and `api.bazaro.online` → Render.

bazaro.online

Domain
Products
Sharing & Transfer
Advanced DNS

DNS TEMPLATES ? Choose DNS Template

HOST RECORDS ?

Actions ▾ Filters ▾ Search 🔍

Type	Host	Value	TTL	
A Record	@	216.11.191.1	Automatic	
CNAME Record	api	bazaro-api-stripe.com.	Automatic	
CNAME Record	www	f21381f22e2d9c9a5b170.com.	Automatic	

+ ADD NEW RECORD

8. Admin Settings – Runtime Controls

Screenshot of the admin settings panel on the live site, showing maintenance mode and payment toggles.

The screenshot displays the Bazzoro Admin Panel interface. At the top, there's a navigation bar with the Bazzoro logo and a sidebar menu containing options like Dashboard, Users, Products, Services, Bookings, Orders, Payments, Categories, Contacts, Reports, and Reviews. The main content area features a 'Dashboard Overview' section with several key performance indicators (KPIs) represented by colored cards:

- PRODUCTS**: 7 items, up 33% vs last month.
- SERVICES**: 7 items, up 8% vs last period.
- ORDERS**: 13 orders, up 11% vs last period.
- REVENUE**: Rs. 5,400, down 98% vs last month.
- CATEGORIES**: 12 categories, down 2% vs last period.
- TOTAL REVIEWS**: 1 review, up 6% vs last period.
- HIDDEN RATIO**: 0.0%, down 1% vs last period.
- AVG RATING**: 5.0 rating, up 4% vs last period.

Below the KPI cards, there are sections for 'Most Active User' (Diani, 1 order) and 'Top Category' (Tutoring Services, 5 listings). A 'Recent Orders' table at the bottom lists three transactions, all of which have been cancelled:

ID	Description	Status	Amount
ig-test-1775842846801-6f018cbb11@example.com	Purchase Order Cancel Buyer to Product Order Cancel Seller	CANCELLED	Rs. 1,250
ig-test-1775842846546-00d4d54f389xa2@example.com	Purchase Order Buyer to Product Order Seller	CANCELLED	Rs. 2,500
ig-test-1775842845347-a35818f3c02da@example.com	Purchase Order Cancel Buyer to Product Order Cancel Seller	CANCELLED	Rs. 1,250

7. Post-Deployment Validation (Smoke Testing)

The following smoke tests were executed immediately after deployment to verify core functionality in the production environment.

#	Test Case	Result	Notes
1	Frontend loads on https://www.bazaro.online	Pass	Home page renders correctly
2	Backend API reachable on https://api.bazaro.online	Pass	Settings endpoint returns 200
3	User registration	Pass	New account created successfully
4	User login / JWT issuance	Pass	Token returned and stored in session
5	Product listing retrieval	Pass	Listings load with pagination
6	Product detail view	Pass	Detail page renders correctly
7	Service booking flow accessible	Pass	Booking form loads correctly
8	Payment path when paymentsEnabled	Pass	Stripe checkout reachable
9	Payment blocked when payments disabled	Pass	Configured message shown to user
10	Maintenance mode toggle from admin settings	Pass	Toggle updates state correctly
11	Non-admin redirected during maintenance mode	Pass	Maintenance page shown
12	Admin can access admin login during maintenance	Pass	Admin bypass works
13	Admin can access admin routes during maintenance	Pass	No redirect for admin
14	Logout confirmation modal appears	Pass	Custom modal renders correctly
15	Maintenance enable confirmation modal	Pass	Custom modal with correct UI
16	API docs accessible on /api-docs	Pass	Swagger UI loads all endpoints

8. Issues Encountered & Resolutions

Issue 1: Direct Frontend Routes Returned 404 on Refresh

Root Cause:	SPA fallback routing was not configured on Vercel, causing direct URL access to return 404.
Fix Applied:	Added SPA rewrite rules in Vercel configuration to redirect all unknown routes to index.html.
Result:	Routes such as /admin/login now load correctly when accessed directly.

Issue 2: Browser Native Confirm Dialogs – Inconsistent Styling

Root Cause:	window.confirm() was used for confirmation dialogs, producing unstyled browser-native popups.
Fix Applied:	Replaced all window.confirm() calls with a custom-built styled confirmation modal component.
Result:	Consistent UX and visual design across all confirmation interactions.

Issue 3: Confirmation Modal Clipping Inside Transformed Container

Root Cause:	The modal was rendered inside a container with CSS transform applied, causing visual clipping.
Fix Applied:	Refactored the modal to render via a React portal targeting document.body.
Result:	Modal now overlays correctly across all views without clipping.

Issue 4: Admin Redirection Issues During Maintenance Mode

Root Cause:	Maintenance mode redirect logic was too aggressive, blocking admin routes as well.
Fix Applied:	Updated routing logic to explicitly allow /admin/login and added admin bypass behaviour.
Result:	Admin users can access required routes while non-admin users are correctly restricted.

Issue 5: Backend Slow After Inactivity (Cold Start)

Root Cause:	Render's free tier spins down inactive web service instances to save resources.
Fix Applied:	Documented the expected cold-start delay. No code change required for this tier.
Result:	System works correctly with an initial wake-up delay of ~30–60 seconds.

9. Rollback & Recovery Plan

Backend Rollback (Render)

- Open the Render dashboard and navigate to the Bazaaro web service.
- Go to the Deployments tab and identify the last known stable deployment.
- Click Redeploy on the stable deployment to promote it to production.
- After rollback, verify the API docs and critical endpoints (login, listings, settings).

Frontend Rollback (Vercel)

- Open the Vercel dashboard and navigate to the Bazaaro project.
- Go to the Deployments tab and identify the last successful production deployment.
- Click Promote to Production on the stable deployment.
- Verify frontend routing and API integration after promotion.

If a Secret or API Key is Compromised

- Immediately rotate the affected key/token in its provider dashboard (Stripe, Twilio, etc.).
- Update the corresponding environment variable in Render and/or Vercel.
- Trigger a fresh deployment on both platforms to apply the new values.
- Audit logs for any unauthorized usage during the exposure window.

Database Recovery (MongoDB Atlas)

- Use MongoDB Atlas automated backups or snapshots to restore data if corruption occurs.
- Select the appropriate point-in-time snapshot from the Atlas backup panel.
- Restore to a new cluster and verify data integrity before pointing the backend at it.
- Run minimal validation tests (user login, listing retrieval) after restore.

10. Final Live References

Resource	URL
GitHub Repository	https://github.com/kiruluchamika/Hyperlocal-Marketplace-for-Products-Services
Live Frontend	https://www.bazaro.online
Live Backend API	https://api.bazaro.online
API Documentation	https://api.bazaro.online/api-docs
Settings Endpoint	https://api.bazaro.online/api/settings
Release Tag	v1.0.0-beta (GitHub Releases)